General video to anime video translation

Presented as a part of IE 643 Course Project:

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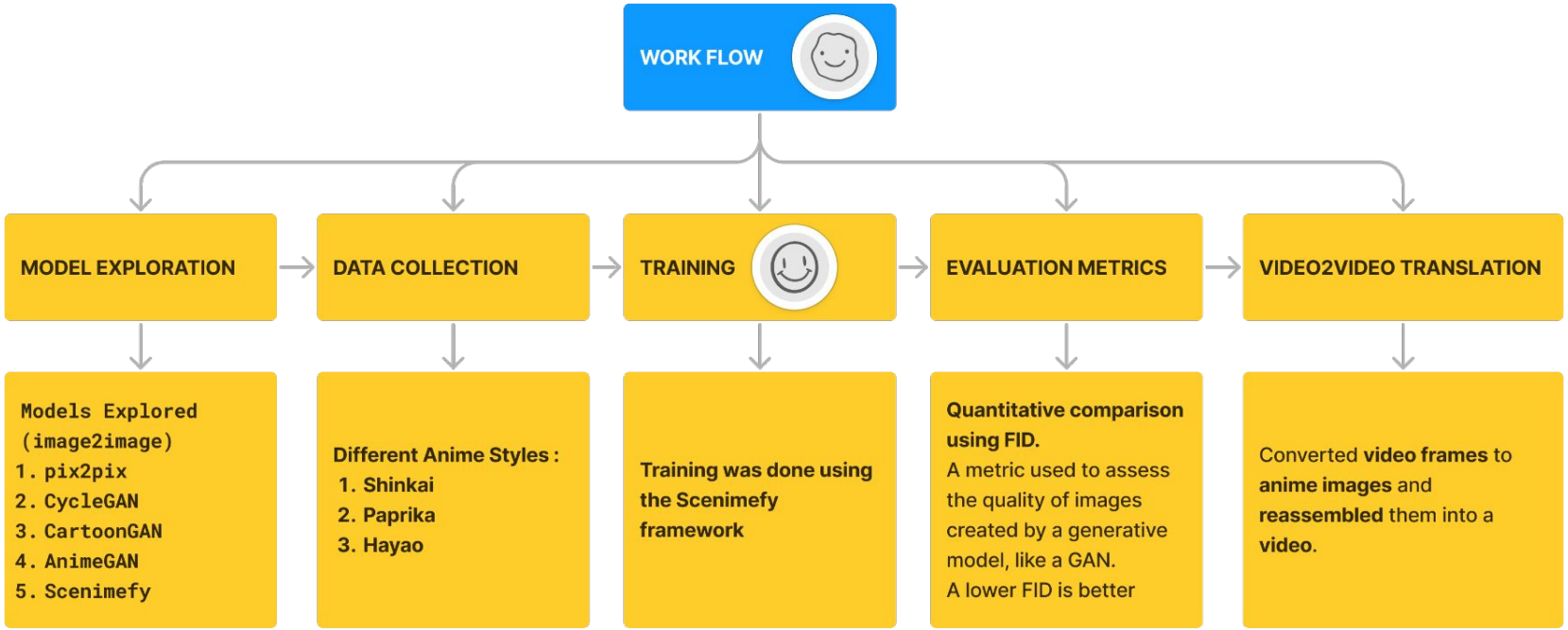
Problem Statement

The goal of this project is to develop a **video-to-anime translation model** that can transform real-life videos into anime-style videos while preserving motion, character identity, and events

This project aims to address the following key challenges:

- **Real-to-Anime Transformation**: Converting complex, photorealistic textures into simplified anime forms while retaining the original structure and dynamics.
- **Temporal Coherence** : Ensuring smooth and seamless transitions between consecutive frames to avoid jitter and flicker artifacts.

Workflow Diagram



Summary of Work Before Prep Review

Data Collection:

- Gathered data sets featuring various anime styles:
  - **Shinkai, Paprika, Hayao**
- Reviewed **existing image-to-image style transfer models**.

Initial Model Selection:

- Tested **CycleGAN**, **AnimeGAN**, and **CartoonGAN**.
- Analyzed model architectures and suitability for video-to-anime conversion..

Custom Model Development:

- Attempted to design a **custom GAN** model tailored for anime style transfer.

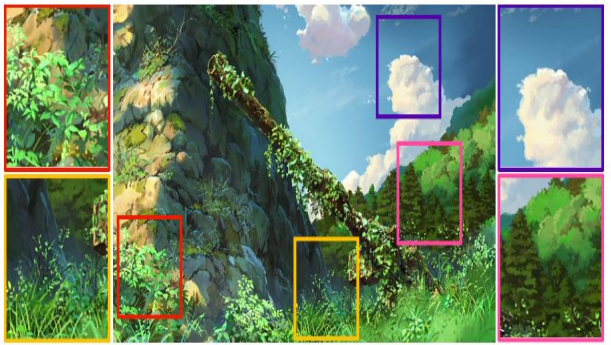


HAYAO (AnimeGAN)

Work Done After Prep Review

Scenimefy Integration :

- Discovered and explored **Scenimefy** model architecture.
- Used **pre-trained Scenimefy model** to convert real images into anime images.
- Utilized libraries to apply Scenimefy for video-to-anime transformations.



Training and Code Development :

- Learned to train Scenimefy from scratch and implemented **checkpoint loading** for model continuity.

Evaluation Metric :

- Implemented **Fréchet Inception Distance (FID)** to assess the quality of anime-style outputs.

Frame Extraction and Anime Conversion Process

- Extracted frames from videos -> Anime Conversion -> Video Reconstruction

Proposed Approach

For the image-to-anime translation, we utilized **Scenimefy**. It has a **three stage pipeline**:

1. Pseudo paired data generation

Source Generator (Gs):

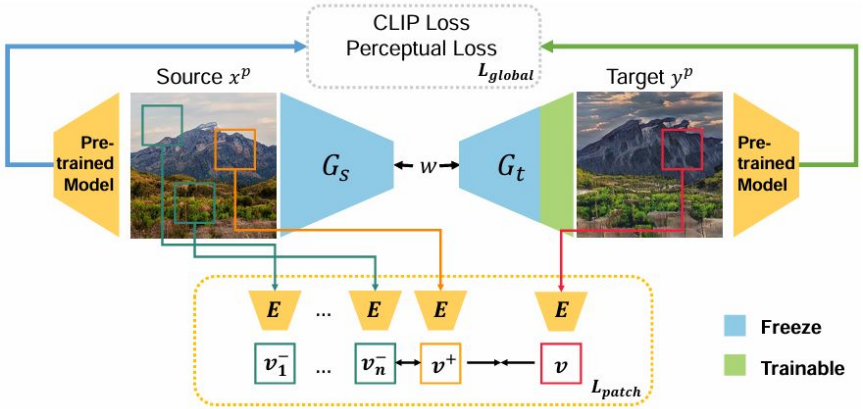
- Pre-trained on a real-scene domain.
- Kept fixed throughout the training process.

Target Generator (Gt):

- Initialized with pre-trained weights.
- Optimized with rich model priors (e.g., CLIP, VGG).
- Early layers frozen to adapt to the anime scene domain.

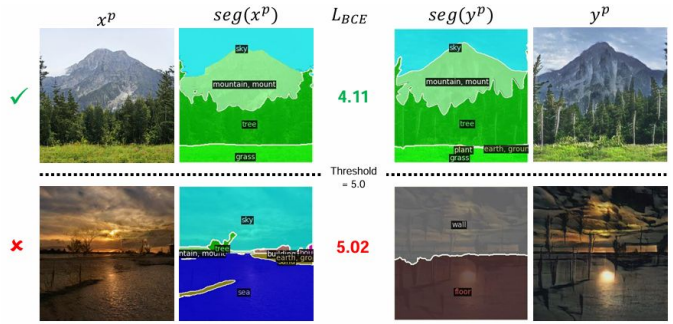
Patch-wise Contrastive Loss:

- Uses CLIP embedders (E) to preserve local spatial details.
- Ensures content integrity and alignment with anime style



Proposed Approach

2. Semantic Segmentation Guided Data Selection



1. **Need for Filtering:**
   - Raw pseudo paired data may lack structural consistency.
   - Semantic segmentation is used to filter out poor-quality samples.
2. **Mask2Former for Filtering**:
   - Mask2Former generalizes well to both real and anime domains.
   - We apply Mask2Former to filter data based on:
     - **Semantic consistency**: Evaluated using pixel-wise cross-entropy loss
     - **Semantic abundance**: Images with only one detected category are excluded (indicates low semantic richness).
3. **Filtering Criteria:**
   - Samples with $\mathbf{L_{BCE}}$ higher than 5.0 are discarded (indicating low consistency).
   - Images lacking sufficient semantic details (only one category) are also filtered out.
4. **Outcome:**
   - Higher quality pseudo paired data is retained.

Proposed Approach

3. Semi supervised image-to-image translation

Framework Structure : Supervised branch & **Unsupervised** branch

1. **Supervised Training Branch** :

- Goal: Map real scene images to anime style via mapping G :X→Y using paired data P = { x_i^P, y_i^P}.
- Conditional GAN Loss (Adversarial Objective):

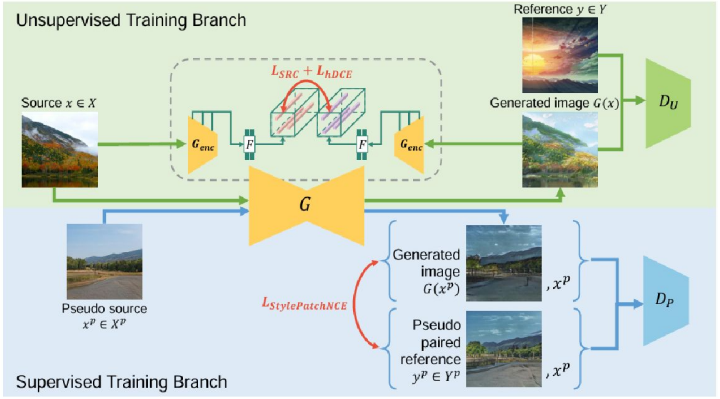
$$L_{cGAN}(G, D_P) = E[y^p, x^p][\log D_P(y^p, x^p)] + E[x^p][\log(1 - D_P(x^p, G(x^p)))]$$

- Patch Discriminator D_P: Distinguishes between true pairs (y^p, x^p) and generated pairs $(G(x^p), x^p)$.

$$L_{global} = D_{CLIP}(x^p, y^p) + \lambda_{lpip} lpips(x^p, y^p)$$

$$L_{patch}(v, v^+, v^-) = -\log\left[\frac{\exp(v \cdot v^+)}{\exp(v \cdot v^+) + \sum_{i=1}^{N} \exp(v \cdot v_i^-)}\right], \quad (2)$$

$$L_{finetune} = L_{GAN}^t(G_t, D) + \lambda_{global} L_{global} + \lambda_{patch} L_{patch},$$



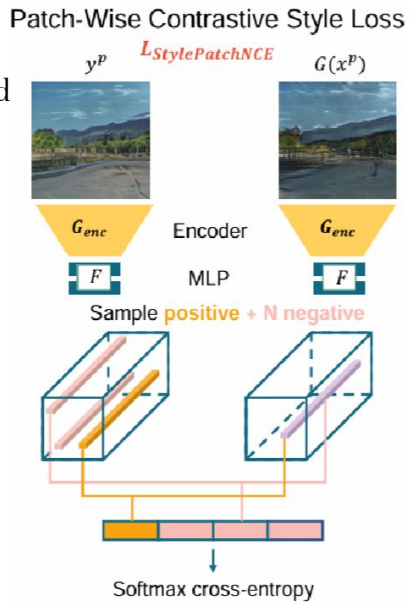
Proposed Approach

Novel StylePatchNCE Loss :

- **Patch-wise contrastive loss** compares patches at the same location in $G(x^p)$ and
  - **Positive patches :** Closer together.
  - **Negative patches :** Far apart.
- **Training objective :**

$$L_{sup} = L_{cGAN}(G, D_P) + \lambda_{style} L_{StylePatchNCE}(G, F, Y^p)$$

- $\mathbf{L_{StylePatchNCE}}$**:** Helps learn fine anime textures and local style similarity.



Outcome:

The supervised branch enhances image translation with **robust local style similarity** and captures fine details using the **patch-wise contrastive loss**.

Proposed Approach

2. Unsupervised Training Branch :

Heterogeneous semantic relations within scene images (e.g., diverse patches like mountain, sea, etc.) were preserved by implementing **Semantic Relation Consistency Loss (LSRC)** and **Hard Negative Contrastive Loss (LhDCE)**

Loss Functions Explained:

- **LSRC**:
  - Minimizes **Jensen-Shannon Divergence (JSD)** between **real image x** and its generated counterpart **G(x)**.
  - **Goal:** Preserve **semantic consistency** between original and stylized patches.
- **LhDCE**:
  - Patch-wise contrastive loss that **increases discriminative difficulty** of negative samples.
  - **Goal:** Enhance model's **discriminative power** for precise stylization

Dataset and Dataset Preprocessing

Datasets for Real-to-Anime Translation

- **LHQ Dataset**:
  - Contains 30,000 nature landscape images, representing real-world scenes.
  - Saved in the unpaired_s2a as train A
- **Anime Dataset**:
  - Comprises 5,958 Shinkai-style anime scene images.
  - Used a json format to identify frame numbers across nine Shinkai movies (1,986 frames total) and then downloaded and renamed each movie as specified from youtube.
  - Extract the frames listed in *json* file, with the option to crop them to adjust the resolution
  - Saved in the directory unpaired_s2a as train B
- **Pseudo-Paired Dataset**:
  - Contains 30,000 synthetic **pseudo-paired images** created using StyleGAN.
  - Saved in the directory : .pair_s2a

Experimental Setup and Details

Model Components and Architecture

- Base model : **Contrastive Unpaired Translation** (CUT)
- Discriminator (netD) : Basic model (70 x 70 PatchGAN)
- Generator (netG) : resnet_9blocks
- Feature extraction layers (netF) : mlp_sample

Training Parameters

- **Batch size** 1 and **image dim** 704 x 704
- Noise Contrastive Estimation Loss applied (**nce_layers**) : 0, 4, 8, 12 & 16
- **Instance Normalization** applied in netD & netG

Loss functions and their weights

- **GAN Loss** (lambda_GAN) : Pushing the generator to produce realistic outputs (set to 1)
- **Hierarchical Density Contrastive Estimation Loss** (lambda_HDCE) : Used to enforce similarity between generated and original images. (also set to 1)
- **Self-Reconstruction Consistency Loss** (SRC) : Helps in maintaining similarity between the input and output images to improve translation quality and note its weight (set to 1).

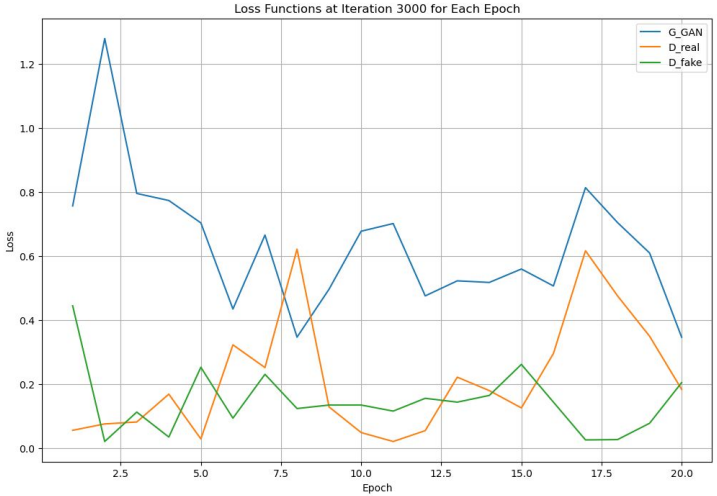
Experimental Setup and Details

Learning Rate Schedule & Momentum Parameters

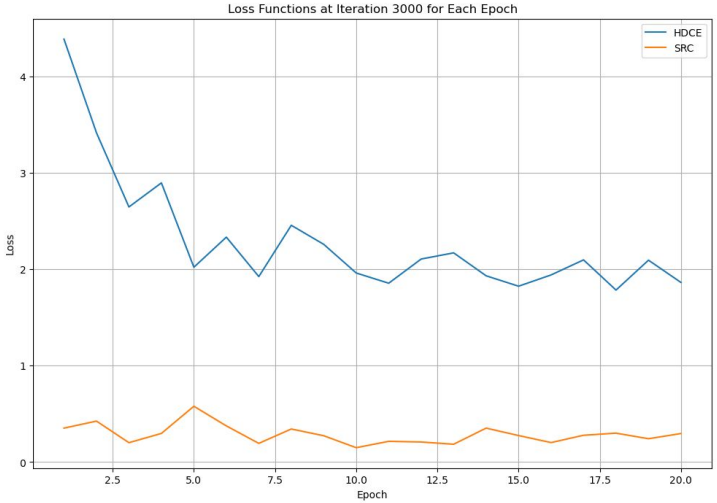
1. **Training Phase (10 Epochs)**
   - Initial learning rate (lr): 0.0002
   - **Constant learning rate** for the first 10 epochs (n_epochs).
2. **Decay Phase (10 Epochs)**
   - Learning rate **linearly decreases** to zero over the next 10 epochs (n_epochs_decay).
   - Enables gradual convergence, refining output quality and stabilizing training.
3. **Momentum Parameters for Adam Optimizer**
   - Beta 1 (beta1): 0.5 – Controls the exponential decay rate for the first moment (mean) of gradients. Lower value gives more stable but slower convergence.
   - Beta 2 (beta2): 0.999 – Controls the decay rate for the second moment (variance). Keeps updates from oscillating while maintaining learning momentum.
4. **End of Training**
   - After 20 total epochs (10 training + 10 decay), the learning rate reaches zero.
   - No further weight updates occur, allowing the model to reach a stable, optimized state.

Experimental Results

Loss Curves



Loss functions : G_GAN , D_real & D_fake



SRC & HDCE loss

Fréchet Inception Distance

1. No of anime images (SummerWar) = 1357

   No of paired images = 3000

   FID score = 361.29



2. No of anime images (Shinkai) = 5958

   No of paired images = 30000

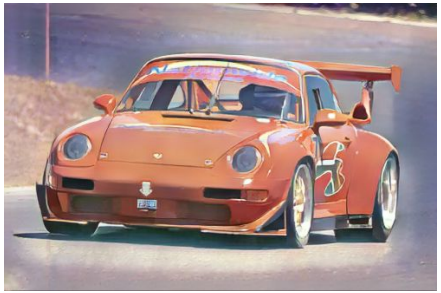
   FID score = 58.104



Generated images and Videos



Original video link
https://drive.google.com/file/d/1kdIK9668HJ2RyuNSGW9nHpVYy4UBVt10/view?usp=sharing



Translated video link
https://drive.google.com/file/d/1kdIK9668HJ2RyuNSGW9nHpVYy4UBVt10/view?usp=sharing

Generated images comparison



Method	LHQ (real)	CartoonGAN	AnimeGAN	Ours
FID↓	121.807	67.200	67.739	**58.104**

Novelty and Future Work

Real-Time Enhancements: Optimize algorithms for faster processing, enabling real video to anime video conversion.

Improving Human Animation: Enhance model performance for human faces and figures by integrating specialized architectures and fine-tuning with human-centric anime datasets.

User Feedback Integration: Implement feedback loops and user studies to gather insights on video quality and usability, focusing on style fidelity and content retention for iterative model improvement.

Conclusion

Problem Addressed

Transforming real videos into anime-styled videos while balancing style fidelity and semantic accuracy.

Proposed Solution

Scenimefy Framework for image to image translation

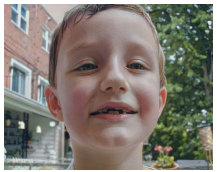
- Three Stages:
  1. Pseudo Paired Data Generation
  2. Semantic Segmentation-Guided Data Selection
  3. Semi-Supervised Image-to-Image Translation

Significance of Results

- Achieved seamless anime-style transformations for high quality videos, by using state-of-the-art baselines.
- Achieved good **texture detail** and **semantic alignment**.

Problems faced

- High computation
- Detail Preservation Challenges
- Semantic translation inconsistencies





Failed to translate these images to anime style

References

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